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Question 0 pts



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Question 1 pts



A researcher compares students attending project-based schools vs. lecture-based schools on creativity scores. What is the independent variable?

- ☐ School type
- ☐ Creativity score
- ☐ Students
- ☐ Teacher experience



Question 1 pts



A **ratio scale** variable has a true zero point, allowing for meaningful multiplication and division of values.

- ☐ True
- ☐ False



Question 1 pts



Ranking cities from most to least livable represents which scale?

- ☐ Ordinal
- ☐ Nominal
- ☐ Interval
- ☐ Ratio



Question 1 pts



A variable measured in years of education is on a ratio scale.

- ☐ True
- ☐ False



Question 2 pts



Match each term on the left with its correct definition or description from the list provided in the dropdown menus. Select the most appropriate option for each term.

Independent Variable

The variable that is manipulated or categorized to see its effect.

Dependent Variable

The variable that is measured as an outcome or response.

Standard Deviation

A summary measure that indicates how spread out scores are around the mean.

Nominal Scale

Categories such as “red,” “blue,” “green,” with no intrinsic order.

Ordinal Scale

Data that show rank order but not equal spacing between ranks.

Histogram

A graph displaying the frequency distribution of a continuous variable.



Question 1 pts



The dependent variable is the one that the researcher manipulates.

- ☐ True
- ☐ False



Question 1 pts



Operationalisation is the process of transforming an abstract concept, like “intelligence” or “age,” into a measurable variable that can be observed or recorded.

- ☐ True
- ☐ False



Question 1 pts



The mean is less sensitive to extreme values than the median.

- ☐ True
- ☐ False



Question 1 pts



The **median** is particularly useful when:

- ☐ The data contain extreme values
- ☐ The data are symmetrically distributed
- ☐ The data are nominal
- ☐ The mean equals the mode



Question 1 pts



In a positively skewed distribution, the mean is typically:

- ☐ Greater than the median
- ☐ Equal to the median
- ☐ Less than the median



Question 1 pts



Suppose you compute the variance of a dataset in R using `var(x)`. What does this value represent?

- ☐ The average squared deviation from the mean.
- ☐ The average deviation from the median.
- ☐ The square root of the standard deviation.
- ☐ The sum of absolute deviations.



Question 1 pts



A 30-item math test was graded using the following procedure:

- A correct response = **+1**
- A blank response = **0**
- An incorrect response = **-1**

The maximum possible test score was 30, and the minimum possible score was -30 .
The standard deviation of the class's test scores was reported as -2.13 .

What conclusion should be drawn from this information?

- ☐ A calculation error was made in determining the standard deviation
- ☐ Some students received negative scores.
- ☐ The test was too hard for this class.
- ☐ Most students received positive scores.
- ☐ The class performed poorly on this test.



Question 1 pts



The **interquartile range (IQR)** measures:

- ☐ The distance between the 25th and 75th percentiles
- ☐ The difference between the maximum and minimum
- ☐ The deviation of each score from the mean



Question 1 pts



A group of 30 introductory statistics students took a 25-item test.

After computing the mean and standard deviation, the instructor found that the standard deviation was **0**.

What does this tell you about the test scores?

- ☐ Everyone correctly answered the same number of items
- ☐ About half of the scores were above the mean
- ☐ A calculation error must have been made in determining the standard deviation



Question 1 pts



Which scenario best illustrates Simpson's Paradox?

- ☐ When trends in subgroups reverse when data are combined
- ☐ When two variables are correlated but not causally related
- ☐ When a random sample produces unusual results
- ☐ When data have high variance but low mean



Question 1 pts



If two datasets have the same mean but very different standard deviations, what does that indicate about the data?



Question 1.5 pts



Assume that the average household income in the United States is \$25,000.

Families whose annual income is less than 1.5 standard deviations below the mean qualify for governmental assistance.

If the variance of household income is \$10,000, would a household earning \$24,500 qualify for assistance?

Show all calculations and explain your reasoning.



Question 1.5 pts



At a large university, the average monthly rent for student housing is \$1,200, with a variance of \$16,000.

Students whose monthly rent is more than 1.5 standard deviations above the mean are eligible for a housing stipend to offset high living costs.

Would a student paying \$1,350 per month qualify for the stipend?

Show all your calculations and explain your reasoning clearly in complete sentences.

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